

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I M. Mary Sharlin (192511330) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Mary Sharlin

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

D. Define cloud computing and Explain any four characteristics.

Definition:-

Cloud computing is the delivery of computing services (Servers, storage, databases, networking, Software etc,...) over the internet on a pay-as-you-use basis.

Characteristics:-

- * On-demand self-service: users can access resources whenever needed without human intervention.
- * Broad network access: Services are available over the internet and can be accessed from different devices.
- * Resource pooling: Resources are shared among multiple users using a multi-tenant model.
- * Rapid elasticity: Resources can be increased or decreased quickly based on demand.

Evolution of Cloud computing.

- * Hardware Evolution: Large mainframe computers replaced by personal computers and powerful servers.

* Networking Evolution: Large mainframe computers were replaced by personal computers and powerful servers.

* Software Evolution: Web-based applications replaced traditional desktop software.

* Cloud Platforms: Virtualization and distributed computing enabled providers like AWS, Azure, Google Cloud to offer scalable cloud services.

2. Role of Server Virtualization in Cloud Computing

Server Virtualization:

It allows multiple virtual machines (VMs) to run on a single physical server, improving resource utilization and reducing hardware costs.

Virtualization	Cloud Computing
Technology that creates virtual machines.	Service model that delivers IT resources over the internet.
Focuses on efficient hardware utilization.	Focuses on providing scalable services to users.
Eg: VMware, Hyper-V	Eg: AWS EC2, Microsoft Azure

Role of Data Centers in Delivering cloud services.

* Data centers are facilities that contain server, storage devices, networking equipment, and cooling systems.

Role :-

- * Store and process user data.
- * Host cloud applications and servers.
- * Ensure high availability and reliability.
- * Provide security, backup and disaster recovery.
- * Support scalable cloud infrastructure.

Differentiate IaaS, PaaS, SaaS and XaaS

Service model	Description	Example
IaaS (Infrastructure as a service)	Provides virtual servers, storage	Amazon, EC2
PaaS (Platform as a Service)	Provides a platform for developing and deploying applications	Google App Engine
SaaS (Software as a Service)	Delivers ready-to-use software over the internet.	Gmail
XaaS (Anything as a Service)	Offers any IT service through the cloud	Microsoft 365